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Models and Tools for Cyber-Physical Systems
Exercise sheet 4
WITH SOLUTIONS

Exercise 1: Reachable states

The composed system

$$\texttt{RailRoadSystem1} = \texttt{Controller1} \parallel \texttt{Train}_\texttt{W} \parallel \texttt{Train}_\texttt{E}$$

(Chapter 3, slides 32-35) has four state variables: `west` and `east`, each of which can take two values, and $\texttt{mode}_W$ and $\texttt{mode}_E$, each of which can take three values. Thus, `RailRoadSystem1` has $2 \cdot 2 \cdot 3 \cdot 3 = 36$ states.
Assume we stop an execution once two trains are on the bridge. Out of these 36 states, only 13 states are reachable.
Enumerate *five* of these reachable states (in the interest of time).

.....Solution sketch .....

We denote each state by listing the values of the variables in the order `west`, `east`, $\texttt{mode}_W$, $\texttt{mode}_E$. We use the abbreviations a, w, b, g, r for the values `away`, `wait`, `bridge`, `green`, `red`, respectively.

The initial state is $ggaa$ (i.e., both `west` and `east` are "`green`" and both $\texttt{mode}_W$ and $\texttt{mode}_E$ are "`away`"). For the transitions, there are four possible cases.

(a) No train arrives. We add a transition back to the initial state (self-loop).

(b) The west train arrives. We add a transition to the (new) state $grwa$, since the traffic light for the east train gets switched to red.

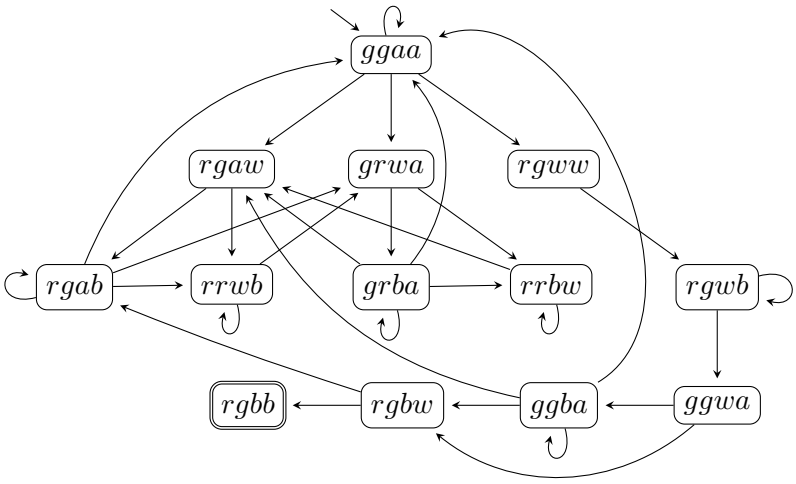
(c) The east train arrives. This case is analogous to case (b), so we add a transition to the (new) state $rgaw$.

(d) Both trains arrive. We add a transition to the (new) state $rgww$ since the controller breaks ties by switching the west traffic light to red.

Now we need to apply the same reasoning to the three newly discovered states, and keep repeating until no new states are found. In the end, the following 13 states are reachable:

$$\{ggaa, rgaw, grwa, rgww, rgab, rrwb, grba, rrbw, rgwb, ggwa, ggba, rgbw, rgbb\}$$

The full reachable part of the transition system is depicted below.



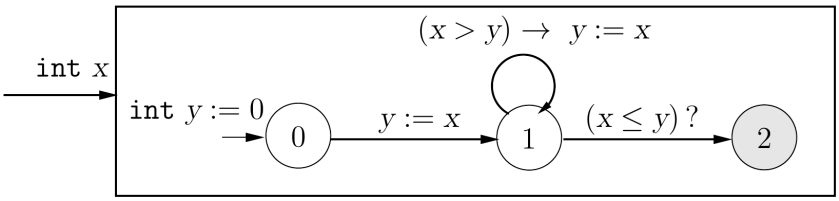
Exercise 2: Monitor

Consider a component `C` with an output variable `x` of type int. Design a safety monitor (given as an extended state machine) to capture the following requirement:

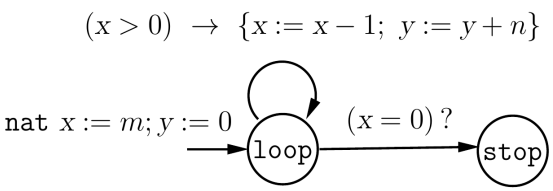
> The sequence of outputs is strictly increasing (i.e., the output in each round (after the first) is strictly greater than the output in the preceding round).

.....Solution sketch .....

The monitor shown below receives the output `x` as input. The monitor maintains a state variable `y` to record the value of the input from the preceding round. In the first round, the monitor only records the first value of `x` and goes to mode 1. In mode 1, the monitor compares the current value of `x` to the value of `y` and then updates `y` to the value of `x` if `x` was larger. The monitor enters mode 2 exactly when the desired safety requirement gets violated.



Exercise 3: Symbolic transition system `Mult(m,n)`
The following extended state machine multiplies two natural numbers m and n:



Describe the transition system symbolically using initialization and transition formulas.
Hint: Recall that there is an implicit variable "mode" here.

.....Solution sketch .....

The transition system `Mult(m,n)` has three state variables: x and y of type nat, and "mode" of the enumerated type $\{\mathsf{loop}, \mathsf{stop}\}$. The initialization formula is given as:

$$x = m \wedge y = 0 \wedge \mathsf{mode} = \mathsf{loop}$$

The transition formula is given as:

$$\begin{array}{ll} & (\mathsf{mode} = \mathsf{loop} \wedge x > 0 \wedge x' = x - 1 \wedge y' = y + n \wedge \mathsf{mode}' = \mathsf{loop}) \\ \vee & (\mathsf{mode} = \mathsf{loop} \wedge x = 0 \wedge x' = x \wedge y' = y \wedge \mathsf{mode}' = \mathsf{stop}) \end{array}$$

Exercise 4: Image computation
Consider the symbolic image computation for a transition system with two integer-valued variables x and y and transitions described by the formula $\varphi_T : x' = 2x \wedge y' = x$. Suppose the region A is described by the formula $0 \leq x \leq 3 \wedge y \leq 5$.

(a) Compute the formula describing the image $\mathrm{Post}(A, \varphi_T)$.

(b) How many states does the image $\mathrm{Post}(A, \varphi_T)$ contain?

.....Solution sketch .....

(a) The conjunction of the given region A and the transition formula is

$$0 \leq x \leq 3 \wedge y \leq 5 \wedge x' = 2x \wedge y' = x.$$

Existential quantification of the unprimed variables yields

$$\exists x, y \colon 0 \leq x \leq 3 \wedge y \leq 5 \wedge x' = 2x \wedge y' = x.$$

First we eliminate the variable x. Because of the conjunct $y' = x$, we can just replace x by y' (and drop the resulting trivial conjunct $y' = y'$):

$$\exists y \colon 0 \leq y' \leq 3 \wedge y \leq 5 \wedge x' = 2y'$$

Next we eliminate the variable y, for which we can just drop the conjunct $y \leq 5$ here (because there clearly exists a value for y such that $y \leq 5$ holds), leading to

$$0 \leq y' \leq 3 \wedge x' = 2y'.$$

Finally, we rename the primed variables to their unprimed counterparts and obtain

$$0 \leq y \leq 3 \wedge x = 2y.$$

(b) The formula allows for four possible values of y. For each of them, the value of x is determined. Thus, the formula describes four different states. In the order (x, y): $(0, 0), (2, 1), (4, 2), (6, 3)$